

(1) EM Spectrum

Ranges	γ	X	UV	vis	IR	μw	radio
Frequency (ν)	↑						↓
Energy (E)	↑						↓
Wavelength (λ)	<0.01	0.01-10	10-400	400-700	700-1e6	1e6-1e9	>1e9

• vis (violet · blue · green · yellow · orange · red)

(2) $c = \lambda \nu$ (Speed of light = wavelength × frequency)

$\lambda \downarrow / \nu \rightarrow$	Hz	kHz	MHz	GHz
m	3.00e8 (m/s)	3.00e5 (m·kHz)	3.00e2 (m·MHz)	3.00e-1 (m·GHz)
nm	3.00e17 (nm·Hz)	3.00e14 (nm·kHz)	3.00e11 (nm·MHz)	3.00e8 (nm·GHz)

(3) $E = h\nu = h(c/\lambda)$ (Energy = Planck × frequency)

$E \downarrow / \lambda, \nu \rightarrow$	λ (m)	λ (nm)	ν (Hz)
J / photon	1.9864e-25 / λ (J·m)	1.9864e-16 / λ (J·nm)	6.626e-34 × ν (J·s)
J / mol	1.1962e-1 / λ (J·m/mol)	1.1962e8 / λ (J·nm/mol)	3.9902e-10 × ν (J·s/mol)
kJ / mol	1.1962e-4 / λ (kJ·m/mol)	1.1962e5 / λ (kJ·nm/mol)	3.9902e-13 × ν (kJ·s/mol)

• Grams → mol → ×6.022e23 for photon count

(4) Quantum Principles

Planck	Energy emitted only in fixed amounts (quanta)
Einstein	Light arrives in packets (photons) of fixed energy
Heisenberg	Position and velocity not both knowable
Bohr	e ⁻ occupy fixed energy levels; explains line spectra

• Orbital = region where an e⁻ is most likely found

• $E = h\nu = h(c/\lambda) = R(1/n_f^2 - 1/n_i^2)$

• $n_i = \sqrt{1/(1/n_f^2 - hc/(R\lambda))}$; $n_f = \sqrt{1/(1/n_i^2 + hc/(R\lambda))}$

Variable	Value	Variable	Value
h	6.626e-34 J·s	h/R	3.0394e-16 s
c	2.9979e8 m/s	hc	1.9864e-25 J·m
R	2.18e-18 J	hc/R	9.1120e-8 m

(5) de Broglie

$$\lambda = h/(mu)$$

λ (m) given $h/(m \cdot u)$; $m \downarrow / u \rightarrow$	m/s	km/s
kg	6.626e-34	6.626e-37
g	6.626e-31	6.626e-34
amu	3.9892e-7	3.9892e-10

• Answer is in m — **x1e9 for nm** (1 nm = 1e-9 m)

(6) Electron Configuration

1. Fill in Aufbau order (below)

2. Respect **s2 p6 d10 f14**

3. Apply the exceptions

• Aufbau = **increasing energy** →: 1s 2s 2p 3s 3p **4s 3d** 4p 5s 4d 5p 6s 4f 5d

Trend	→ across period	↓ down group
Z_{eff}	increases	~constant
Atomic radius	decreases	increases
Ionization energy	increases	decreases
Electron affinity	more favorable	less favorable

• **Cr = [Ar]4s¹3d⁵ · Cu = [Ar]4s¹3d¹⁰** (half/full d wins)
• Impossible if any subshell is over-filled (3s³, 2p⁸)
• Faster than writing Aufbau: read it off the table — s-block (gp 1-2) → ns, p-block (13-18) → np, d-block → (n-1)d; the period number is n
• Can be a transition metal; ground state only

(7) Reading an Electron Configuration

Paramagnetic vs diamagnetic

1. Draw boxes for the outer subshell — **s 1 · p 3 · d 5 · f 7**

2. Fill singly, parallel spins, before pairing (Hund)

3. Count unpaired e⁻

• Any unpaired → **paramagnetic** · all paired → **diamagnetic**

• Full subshells (s², p⁶, d¹⁰) are always paired

Excited vs ground state

• **Ground** = strict Aufbau order, no gaps

• **Excited** = an e⁻ sits higher than Aufbau requires, leaving a **gap below it**; same total e⁻ count

Valence electrons

• Count e⁻ in the **highest** n (outer s + p) = the group number, 1A→1 ... 8A→8

• Everything below the highest n is **core**

(8) Ion Configuration

1. Start from the neutral atom

2. **Anion**: add e⁻ into the outer p until it reaches a noble-gas configuration

3. **Cation**: remove from the **highest** n **first**

• Transition metals lose **4s before 3d**

• Exam ions are main-group, not transition metals

(9) Quantum Numbers, Penetration & Shielding

sublevel = subshell	s	p	d	f
l	0	1	2	3
m_l	0	-1...+1	-2...+2	-3...+3
orbitals ($2l+1$)	1	3	5	7
max e ⁻ (2 per orbital)	2	6	10	14

Symbol	Allowed	Meaning
n	1, 2, 3, ...	Principal energy level, shell (row #)
m_s	±1/2	spin , 2 e ⁻ per orbital (Pauli)

• **energy** set by n and l **only** — m_l , m_s are degenerate

• **energy: s < p < d < f** (within shell, same n) · across shells use **Aufbau** (6)

• $l \leq n-1$, so **3f and 2d don't exist** — check the sublevel is real before ranking it

• Shell n holds n^2 orbitals, $2n^2$ e⁻

• **Penetration** — orbital reaching into core region is less shielded, so **lower energy**.

• **Pauli exclusion** — no two e⁻ carry the same four quantum numbers → 2 per orbital, opposite spins

• **Shielding** — only **core** (inner) e⁻ shield effectively; valence e⁻ shield each other poorly

(10) Z_{eff} & Core Electrons

$$Z_{eff} \approx Z - (\text{core } e^-)$$

1. Core e⁻ = total e⁻ - valence e⁻ (= the **previous noble gas**)

2. $Z_{eff} \approx Z - \text{core } e^-$

• Same period → same core, so the **rightmost** element wins

• Poor valence shielding is why Z_{eff} rises across a period

(11) Rank Atomic Size

• Down a group **bigger** (new shell) · across a period **smaller** (Z_{eff} pulls in)

• ↑ = **lower-left** · ↓ = **upper-right**

• Group position beats period position when both apply

• cation < neutral < anion

(12) Rank Ionization Energy

• Across a period ↑ · down a group ↓ (opposite of radius)

• ↑ = **upper-right** (He ↑ of all) · ↓ = **lower-left**

• $IE_2 > IE_1$ always

• **Big jump** appears once the first **core** e⁻ is removed

(13) Rank Electron Affinity

- **Most eager for an e⁻ = halogens, upper right** — this course writes it both as "most negative *EA*" and "largest positive *EA*"; either phrasing means the same element
- Noble gases and full/half-full subshells ≈ 0 or unfavorable
- ***EA* (X) and *IE* of X are the same magnitude** — F + e⁻ releases exactly what F → F costs

(14) Families & Diagonal Rule

1A	alkali metals	most reactive metals; ↑ down
2A	alkaline earth	reactive, less than 1A
7A	halogens	reactive nonmetals; ↓ down
8A	noble gases	inert — full octet

- **Diagonal rule** — an element resembles its down-right neighbor: Li~Mg · Be~Al · B~Si

HW-only gaps — off the exam list

Oxides — metal oxide = **basic** (K₂O, MgO, Fe₂O₃, Cr₂O₃) · nonmetal oxide = **acidic** (P₄O₁₀, SO₂, CO₂, Cl₂O₇) · Al₂O₃ and BeO = **amphoteric**

Formula from charges — 1A 1+ · 2A 2+ · 3A 3+ · 5A 3- · 6A 2- · 7A 1-; criss-cross and reduce

- Same-group swap keeps the pattern: BaS → **CaO**

- 2A + 3F₂ → 2AF₃ ⇒ A is 3+ (**Al**) · 3Li + Z → Li₃Z ⇒ Z is 3-, so Mg gives **Mg₃Z₂**

Bond-type ID — ionic **and** covalent together = a salt of a **polyatomic ion** (NH₄NO₃, K₂SO₄); all-nonmetal = covalent only

Family reactions

- 2K(s) + 2H₂O(l) → 2KOH(aq) + H₂(g)
- H₂(g) + Cl₂(g) → 2HCl(g)

Odds & ends

- Lanthanides = 4f row (**Ce**) · actinides = 5f (U)
- Diatomic: H₂ N₂ O₂ F₂ Cl₂ Br₂ I₂
- Metallic character ↑₂ down, ↓₂ across → **lower-left** wins
- Isolelectronic = matches a noble gas e⁻ count (K⁺, Cl⁻, Ca²⁺ = Ar)

(15) Lattice Energy

Energy **released** when gaseous ions form the solid crystal.

$$U \propto \frac{q_1 q_2}{r_1 + r_2}$$

1. Compare charges first

2. If charges tie, smaller ions win

- ↑ charge → ↑ **U** (dominant) · ↑ ionic radius → ↓ **U**

(16) Electronegativity

EN = an atom's ability to attract the electrons shared in a bond.

$$\Delta EN = |EN_A - EN_B|$$

- Read *EN* off the chart — *EN* ↑ up and to the right, F highest, noble gases excluded

(17) Bond Polarity

1. Take the absolute difference (ΔEN)

2. Classify

~0	nonpolar covalent
0.4 – 1.9	polar covalent
≥ 2.0	ionic

- **"Most polar"** → ΔEN **alone** (largest difference wins)
- **"Strongest / shortest bond"** → **bond order FIRST**: triple > double > single. ΔEN only breaks ties between equal orders — a C≡N beats a high- ΔEN Si=O
- Larger ΔEN → stronger, shorter bond, but the guide flags this as *not always true*

(18) Formal Charge & Resonance

$$FC = (\text{valence } e^-) - [\text{nonbonding } e^- + \# \text{ bonds}]$$

1. *FC* on every atom

2. Check ΣFC = the overall charge

3. Pick the structure with *FC*'s **nearest 0**

4. Put any -1 on the **more electronegative** atom

- Equivalent resonance forms → real bonds are the **average** (e.g. 1½)

(19) Lewis Structures x4

1. Total valence e⁻ — +1 per negative charge, -1 per positive
2. Central atom = least electronegative, never H
3. Single bonds out to every atom
4. Fill outer octets with lone pairs (H takes 2)
5. Leftover e⁻ onto the central atom
6. Central below 8 → pull an outer lone pair into a double/triple bond, **unless** Al/B/Be
7. Check formal charges — see (18)

Four structures — one of each type:

(a) Normal octet	8 valence e ⁻ per atom (H wants 2) — H ₂ O, CH ₄ , NH ₃ , CO ₂ , CH ₂ Cl ₂ , CH ₃ CH ₂ Cl
(b) Incomplete octet	Al, B, Be only — content below 8, do not add double bonds to fix. BF ₃ /BCl ₃ /AlI ₃ leave 6 e ⁻ on the center; BeCl ₂ leaves 4
(c) Expanded octet	Central atom period 3 or below (S, P, Cl, Xe, As, I) uses empty d orbitals for >8 — AsCl ₅ , XeI ₂ , SO ₄ ²⁻
(d) Organic	-OH alcohol · C=O aldehyde/ketone · -COOH carboxylic acid · -NH₂ amine

- Every C makes 4 bonds; a -COOH carbon carries one **=O** and one **-O-H**; every O gets 2 lone pairs
- Ions go in brackets with the charge outside
- If the all-single-bond form leaves the center with a nonzero *FC*, add double bonds to zero it (e.g. two S=O in SO₄²⁻)

(20) Photoelectric Effect (*hν* - binding / work)

$$KE = h\nu - \Phi = h(c/\lambda) - \Phi = \frac{1}{2}m_e u^2$$

$$u = \sqrt{(2hc/m_e)(1/\lambda - 1/\lambda_{thr})}$$

Variable	Value
<i>h</i>	6.626e-34 J·s
<i>m_e</i>	9.11e-31 kg
<i>hc</i>	1.9864e-25 J·m
2 <i>hc/m_e</i>	4.3609e5 m ³ /s ²

Have	Want	Use
ν_{thr} or λ_{thr}	Φ	$\Phi = h\nu_{thr} = h(c/\lambda_{thr})$
Φ	λ_{thr} (max ej. λ)	$\lambda_{thr} = hc/\Phi$
Φ	Φ per mol	$\Phi \times N_A$ (/1e3 for kJ/mol)
Φ, λ	<i>u</i>	1. $KE = h(c/\lambda) - \Phi$ 2. $u = \sqrt{2KE/m_e}$
λ, λ_{thr}	<i>u</i>	2nd eq above (skips Φ and <i>KE</i>)

(21) Heisenberg Uncertainty

$$\Delta x \cdot \Delta p \geq \frac{h}{4\pi}, \quad \Delta p = m\Delta u$$

1. $\Delta p = m \cdot \Delta u$ (kg, m/s)

2. $\Delta x \geq h/(4\pi \cdot \Delta p)$

- Result is the **minimum** uncertainty
- Rearrange the same way to solve for Δu
- Keep everything in kg, m, s

(22) Born-Haber Cycle

$$\Delta H^\circ_f = \Delta H_{sub} + \Sigma IE + \frac{1}{2}BE + EA + U$$

1. List every step with its sign: sublimation (+), *IE* (+), bond dissociation (+), *EA* (usually -), *U* (-)
2. **Halve** the bond dissociation if you need only one atom
3. Chain: elements → gaseous atoms → gaseous ions → solid
4. Set the chain equal to ΔH°_f
5. Solve for the single unknown
 - Use *IE₂* as well when the cation is 2+
 - *U* runs ions(g) → solid, so it comes out **negative**; report |*U*| if asked for a positive lattice energy

(23) Bond Enthalpy

$$\Delta H_{rxn} = \Sigma \Delta H(\text{bonds broken}) - \Sigma \Delta H(\text{bonds formed})$$

1. List bonds broken (reactants) and formed (products), using the balanced coefficients
2. Σ BE(broken), always positive
3. Σ BE(formed), subtracted
4. $\Delta H = \Sigma$ broken - Σ formed
 - Bonds unchanged on both sides cancel — skip them
 - Negative ΔH = exothermic
 - For "heat released by *g* of X," convert *g* → mol and scale